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13.11 Numerical Use of the Sampling Theorem

We have met the sampling theorem before, in §4.5 (in relation to the accuracy of the trapezoidal rule for integration); in §6.9, where we implemented an approximating formula for Dawson’s integral due to Rybicki, and in §12.1, where we first saw it in a Fourier context. Now that we have become Fourier sophisticates, we can readily supply a derivation of the formula in §6.9 and illustrate the use of the sampling theorem as a purely numerical tool. Our discussion is identical to Rybicki [1].

For our present purposes, the sampling theorem is most conveniently stated as follows: Consider an arbitrary function $g(t)$ and the grid of sampling points $t_n = \alpha + nh$, where n ranges over the integers and α is a constant that allows an arbitrary shift of the sampling grid. We then write

$$g(t) = \sum_{n=-\infty}^{\infty} g(t_n) \operatorname{sinc} \frac{\pi}{h}(t - t_n) + e(t) \quad (13.11.1)$$

where $\operatorname{sinc} x \equiv \sin x/x$. The summation over the sampling points is called the *sampling representation* of $g(t)$, and $e(t)$ is its error term. The sampling theorem asserts that the sampling representation is exact, that is, $e(t) \equiv 0$, if the Fourier transform of $g(t)$,

$$G(\omega) = \int_{-\infty}^{\infty} g(t) e^{i\omega t} dt \quad (13.11.2)$$

vanishes identically for $|\omega| \geq \pi/h$.

When can sampling representations be used to advantage for the approximate numerical computation of functions? In order that the error term be small, the Fourier transform $G(\omega)$ must be sufficiently small for $|\omega| \geq \pi/h$. On the other hand, in order for the summation in (13.11.1) to be approximated by a reasonably small number of terms, the function $g(t)$ itself should be very small outside of a fairly limited range of values of t . Thus we are led to two conditions to be satisfied in order that (13.11.1) be useful numerically: Both the function $g(t)$ and its Fourier transform $G(\omega)$ must rapidly approach zero for large values of their respective arguments.

Unfortunately, these two conditions are mutually antagonistic — the Uncertainty Principle in quantum mechanics. There exist strict limits on how rapidly the simultaneous approach to zero can be in both arguments. According to a theorem of Hardy [2], if $g(t) = O(e^{-t^2})$ as $|t| \rightarrow \infty$ and $G(\omega) = O(e^{-\omega^2/4})$ as $|\omega| \rightarrow \infty$, then $g(t) \equiv C e^{-t^2}$, where C is a constant. This can be interpreted as saying that, of all functions, the Gaussian is the most rapidly decaying in both t and ω , and in this sense is the “best” function to be expressed numerically as a sampling representation.

Let us then write for the Gaussian $g(t) = e^{-t^2}$,

$$e^{-t^2} = \sum_{n=-\infty}^{\infty} e^{-t_n^2} \operatorname{sinc} \frac{\pi}{h}(t - t_n) + e(t) \quad (13.11.3)$$

The error $e(t)$ depends on the parameters h and α as well as on t , but it is sufficient for the present purposes to state the bound,

$$|e(t)| < e^{-(\pi/2h)^2} \quad (13.11.4)$$

which can be understood simply as the order of magnitude of the Fourier transform of the Gaussian at the point where it “spills over” into the region $|\omega| > \pi/h$.

When the summation in (13.11.3) is approximated by one with finite limits, say from $N_0 - N$ to $N_0 + N$, where N_0 is the integer nearest to $-\alpha/h$, there is a further truncation error. However, if N is chosen so that $N > \pi/(2h^2)$, the truncation error in the summation is less than the bound given by (13.11.4), and, since this bound is an overestimate, we shall continue to use it for (13.11.3) as well. The truncated summation gives a remarkably accurate representation for the Gaussian even for moderate values of N . For example, $|e(t)| < 5 \times 10^{-5}$ for $h = 1/2$ and $N = 7$; $|e(t)| < 2 \times 10^{-10}$ for $h = 1/3$ and $N = 15$; and $|e(t)| < 7 \times 10^{-18}$ for $h = 1/4$ and $N = 25$.

One may ask, what is the point of such a numerical representation for the Gaussian, which can be computed so easily and quickly as an exponential? The answer is that many transcendental functions can be expressed as an integral involving the Gaussian, and by substituting (13.11.3) one can often find excellent approximations to the integrals as a sum over elementary functions.

Let us consider as an example the function $w(z)$ of the complex variable $z = x + iy$, related to the complex error function by

$$w(z) = e^{-z^2} \operatorname{erfc}(-iz) \quad (13.11.5)$$

having the integral representation

$$w(z) = \frac{1}{\pi i} \int_C \frac{e^{-t^2}}{t - z} dt \quad (13.11.6)$$

where the contour C extends from $-\infty$ to ∞ , passing below z (see, e.g., [3]). Many methods exist for the evaluation of this function (e.g., [4]). Substituting the sampling representation (13.11.3) into (13.11.6) and performing the resulting elementary contour integrals, we obtain

$$w(z) \approx \frac{1}{\pi i} \sum_{n=-\infty}^{\infty} h e^{-t_n^2} \frac{1 - (-1)^n e^{-\pi i(\alpha - z)/h}}{t_n - z} \quad (13.11.7)$$

where we now omit the error term. One should note that there is no singularity as $z \rightarrow t_m$ for some $n = m$, but a special treatment of the m th term will be required in this case (for example, by power series expansion).

An alternative form of equation (13.11.7) can be found by expressing the complex exponential in (13.11.7) in terms of trigonometric functions and using the sampling representation (13.11.3) with z replacing t . This yields

$$w(z) \approx e^{-z^2} + \frac{1}{\pi i} \sum_{n=-\infty}^{\infty} h e^{-t_n^2} \frac{1 - (-1)^n \cos \pi(\alpha - z)/h}{t_n - z} \quad (13.11.8)$$

This form is particularly useful in obtaining $\operatorname{Re} w(z)$ when $|y| \ll 1$. Note that in evaluating (13.11.7) the complex exponential inside the summation is a constant and needs to be evaluated only once; a similar comment holds for the cosine in (13.11.8).

There are a variety of formulas that can now be derived from either equation (13.11.7) or (13.11.8) by choosing particular values of α . Eight interesting choices are $\alpha = 0, x, iy$, or z , plus the values obtained by adding $h/2$ to each of these. Since the error bound (13.11.3) assumed a real value of α , the choices involving a complex α are useful only if the imaginary part of z is not too large. This is not the place to catalog all 16 possible formulas, and we give only two particular cases that show some of the important features.

First of all let $\alpha = 0$ in equation (13.11.8), which yields,

$$w(z) \approx e^{-z^2} + \frac{1}{\pi i} \sum_{n=-\infty}^{\infty} h e^{-(nh)^2} \frac{1 - (-1)^n \cos(\pi z/h)}{nh - z} \quad (13.11.9)$$

This approximation is good over the entire z -plane. As stated previously, one has to treat the case where one denominator becomes small by expansion in a power series. Formulas for

the case $\alpha = 0$ were discussed briefly in [5]. They are similar, but not identical, to formulas derived by Chiarella and Reichel [6], using the method of Goodwin [7].

Next, let $\alpha = z$ in (13.11.7), which yields

$$w(z) \approx e^{-z^2} - \frac{2}{\pi i} \sum_{n \text{ odd}} \frac{e^{-(z-nh)^2}}{n} \quad (13.11.10)$$

the sum being over all odd integers (positive and negative). Note that we have made the substitution $n \rightarrow -n$ in the summation. This formula is simpler than (13.11.9) and contains half the number of terms, but its error is worse if y is large. Equation (13.11.10) is the source of the approximation formula (6.9.3) for Dawson's integral, used in §6.9.

CITED REFERENCES AND FURTHER READING:

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